

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	2 October 2026
Team ID	SWTID-2026-5622
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural framework and information as per Table 1 & Table 2.

Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party APIs etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1: Technology Components

S.No	Component	Description	Technology
1	User Interface	ServiceNow Portal / Native Incident Form UI for user interaction and field inputs	HTML, CSS, JavaScript, ServiceNow UI Page / Workspace
2	Application Logic-1	UI Policy Logic (Client-side dynamic visibility, mandatory control, and read-only field conditions)	ServiceNow UI Policy Engine / JavaScript
3	Application Logic-2	Client Script Execution (onSubmit, onChange, onLoad real-time form validation)	ServiceNow Client Scripts (g_form, g_user)
4	Application Logic-3	Business Rules & Data Integrity Validation on Incident Table	ServiceNow Server-Side Scripting / JavaScript
5	Database	Relational storage for Incident records, audit logs, and configuration tables	ServiceNow Relational Database (MariaDB based)
6	Cloud Database	Cloud-hosted ServiceNow High Availability Datacenter Instance	ServiceNow Cloud Infrastructure
7	File Storage	Attachment handling and media storage attached to Incident tickets	ServiceNow Sys_attachment Storage / Cloud File Store
8	External API-1	REST / SOAP APIs for third-party Incident integration and event logging	REST Web Services / JSON / XML
9	External API-2	Single Sign-On (SSO) & User Authentication Interface	OAuth 2.0 / SAML 2.0 / Azure AD
10	Machine Learning Model	ServiceNow Predictive Intelligence for automatic incident categorization (if applicable)	ServiceNow AI / ML Engine
11	Infrastructure (Server / Cloud)	SaaS Deployment on Multi-tenant Cloud Platform	ServiceNow Enterprise Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Client-side JavaScript libraries and ServiceNow web framework components used for UI presentation.	JavaScript / Glide API
2	Security Implementations	Role-based Access Control (RBAC), Access Control Lists (ACLs), SSL/TLS Encryption, and secure Client Script validations.	ACLs, TLS 1.3, HTTPS, Role Authorization
3	Scalable Architecture	ServiceNow Multi-instance cloud architecture supporting high volumes of incident records and concurrent users.	Multi-Instance SaaS Architecture
4	Availability	High-availability cloud datacenters with automatic failover and load balancing ensuring uninterrupted service.	ServiceNow Advanced High Availability (AHA)
5	Performance	Optimized client-side scripts running efficiently without slowing down browser rendering or form submission.	Asynchronous GlideAjax, Client Caching